

ComicVerseAI

AI-Powered Automated Comic Story & Visual Content Generation Platform

Project Name: ComicVerseAI
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CUSTOMER JOURNEY MAP

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	Stage 1: Input & Ideation	Stage 2: AI Generation & Preview	Stage 3: Editing & Export
Actions What does the customer do?	- Enters story theme or concept prompt. - Selects artistic comic style (e.g., Manga, Western Superhero). - Chooses panel count and character preferences.	- Reviews generated AI script and panel breakdown. - Previews synthesized Stable Diffusion visual panels. - Evaluates character visual consistency.	- Customizes dialogue text and speech bubble positions. - Regenerates specific unsatisfied comic panels. - Exports final comic book in PDF format.
Touchpoint What part of the service do they interact with?	- Interactive Web Input Dashboard - Style Selection Carousel - Prompt Configuration Form	- FastAPI Processing Status Bar - Interactive Panel Grid Previewer - Gemini Script Viewer	- Inline Canvas Dialogue Editor - Panel Regeneration Button - PDF Download / Export Module
Customer Thought What is the customer thinking?	"Will the AI understand my complex story plot?" "I hope I don't need technical prompt engineering skills."	"The panel artwork matches my script remarkably well!" "Can I adjust the dialogue in panel 2?"	"This comic layout looks ready for publication." "The downloadable PDF quality is excellent."
Customer Feeling What is the customer feeling?	- Curious and hopeful. - Eager to visualize creative ideas quickly.	- Impressed by visual speed. - Slightly cautious about panel detail accuracy.	- Empowered and satisfied. - Confident to share digital comic with audience.
Process Ownership Who is in the lead on this?	Frontend UX Lead (S Prathistha)	AI Core & Backend Lead (P V Poornima & R Sandhiya)	Export Engine & Integration Lead (Kiruthika K)
Opportunities How can we improve this stage?	Provide pre-built prompt templates for genre starters.	Implement Seed locking for 100% character facial consistency.	Add direct comic sharing to social media platforms.

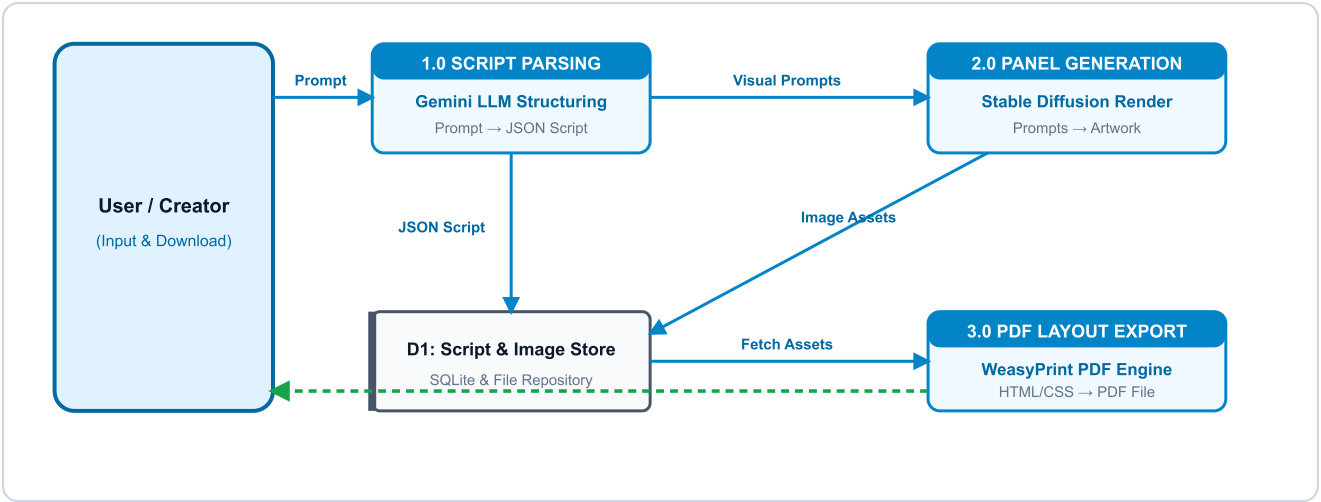
DATA FLOW DIAGRAM (DFD)

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., User, Stable Diffusion API).
Rectangle (Numbered)	Process	An activity that transforms incoming data into outgoing data (e.g., Parse Script, Render Panels).
Double Line / Solid Fill	Data Store	A place where data is held for later use (e.g., Script Database, Panel Image Store).
Labeled Arrow	Data Flow	The movement of data between entities, processes, and stores labeled with data details.

ComicVerseAI Data Flow Diagram Model



SOLUTION REQUIREMENTS

Comprehensive list of requirements defining exactly what the ComicVerseAI solution will do and how well it performs.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Simple session authentication allowing users to manage script projects without complex registration.	Medium
2	Authorization Levels	Creator role access with rights to generate, edit, delete, and download comic projects.	Medium
3	External Interfaces	Integration with Google Gemini API for story script parsing and Stable Diffusion API for image panel rendering.	High
4	Transaction Processing	Asynchronous request handling for AI generation tasks with live progress indicators.	High
5	Reporting & Output	Automated compilation of text script and images into downloadable, high-resolution PDF comics.	High
6	Business Rules	Strict limit of 4 to 8 panels per comic generation run to ensure consistent page layout formatting.	High
7	Compliance	Strict content safety filtering via Gemini safety settings to prevent unsafe prompts or generated imagery.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Criteria
1	Performance & Speed	Rapid story script breakdown and fast image rendering response time.	Script parsing < 3s; Panel generation < 8s per frame.
2	Scalability	System handles concurrent comic generation requests asynchronously.	Supports up to 50 concurrent rendering sessions.
3	Security & Privacy	Secure environment variables management for API keys and input isolation.	Zero exposure of third-party API credentials in client code.
4	Reliability	Graceful fallback handling if third-party image APIs timeout.	99% successful prompt-to-PDF completion rate.
5	Usability	Clean, single-page workflow interface accessible without manual design training.	Complete story-to-PDF creation in under 3 user clicks.

TECHNOLOGY STACK

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the ComicVerseAI solution.

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript (ES6)	Provides a lightweight, responsive UI allowing instant panel previewing and interactive speech bubble editing without complex client build tools.
2	Backend / Server-Side	Python / FastAPI	Offers high-performance asynchronous request handling, essential for orchestrating concurrent AI text and image API calls efficiently.
3	Database / Storage	SQLite / Local File Store	Lightweight local storage for preserving user session prompts, parsed JSON script structures, and generated image asset files.
4	Hosting / Deployment	Uvicorn / Docker	Ensures smooth containerized deployment and rapid local or cloud execution with simple dependency setup.
5	Version Control	GitHub	Enables smooth collaboration between team members, source code version tracking, and structured project repository management.
6	AI Model APIs	Google Gemini Flash API	Translates raw story descriptions into structured JSON comic panel scripts with dialogue and visual prompts.
7	Image Engine API	Stable Diffusion API	Renders high-quality multi-panel comic artwork matching the detailed visual descriptions from Gemini.
8	PDF Export Engine	WeasyPrint / ReportLab	Compiles HTML/CSS panel templates into high-resolution printable PDF documents.